

Rectangle Area

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Problem Understanding

Basically we are given 2 rectangles and we need to find area covered by them.

$$\text{ans} = \text{area1} + \text{area2} - \text{common area}$$

Example Variables

Rect 1

- $ax1 = -3$, $ay1 = 0$
- $ax2 = 3$, $ay2 = 4$

Rect 2

- $bx1 = 0$, $by1 = -1$
- $bx2 = 9$, $by2 = 2$

Area Calculations

So basically:

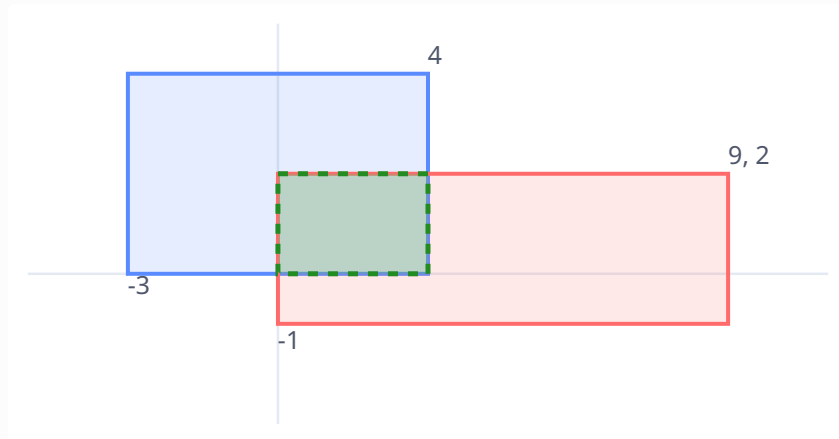
$$\text{abs}(ax2 - ax1) * \text{abs}(ay2 - ay1)$$
$$(6 \times 4) = 24$$

And:

$$\text{abs}(bx2 - bx1) * \text{abs}(by2 - by1)$$
$$(9 \times 3) = 27$$

Common Area Analysis

And for common area:



Obs:

- For x axis: $(\max(ax1, bx1) - \min(ax2, bx2))$
- For y axis: $\text{abs}(\max(ay1, by1) - \min(ay2, by2))$

Implementation Code

```
class Solution {
public:
    int computeArea(int ax1, int ay1, int ax2, int ay2, int bx1, int by1,
int bx2, int by2) {

        int ans=0;

        ans+=abs(ax2-ax1)*abs(ay2-ay1);
        ans+=abs(bx2-bx1)*abs(by2-by1);

        ans-=max(0,min(ax2,bx2)-max(ax1,bx1))*max(0,min(ay2,by2)-
max(ay1,by1)); //notice the outer max its there just to ensure that
overlapping width and overlapping height are finite and positive, otherwise
there wont be any overlapping at all

        return ans;

    }
};
```